

Week 10

Keel Bone Chicken Project

03/09/26



College of Engineering

This Week

Weekly Progress

- **PCB Updates**
 - Appointment with DL-Engineering
 - Layout Updates
 - Design Changes
- **IEEE Paper**
 - General Guidelines
 - Documentation for Project
- **Agriculture Updates**
 - Data working properly
 - CSV saves with intervals

PCB Updates

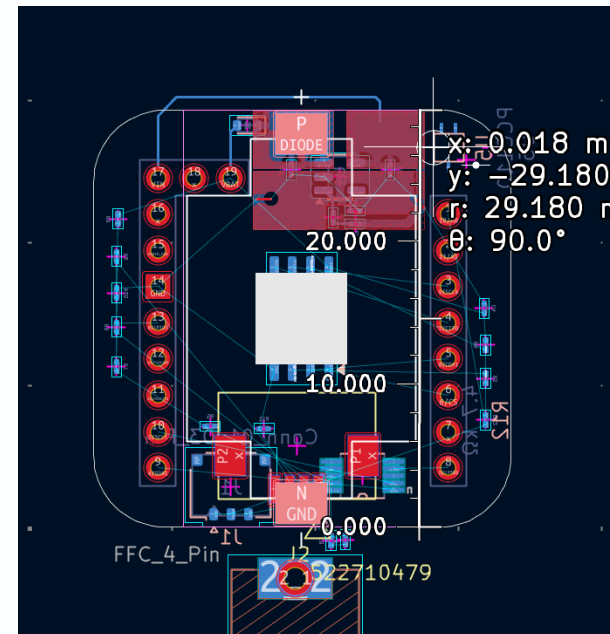
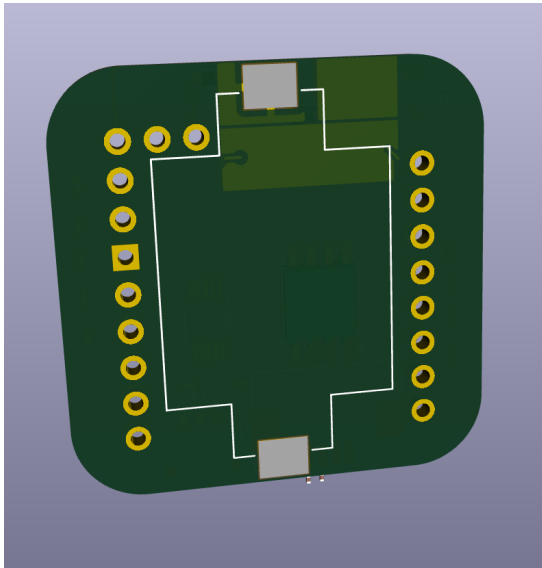
Appointment with DL-Engineering

- Meeting today at 2:30 PM
- Underlying questions:
 - Assembled with BOM or bare board?
 - Number of PCBs assembled?
- Contact for DL-Engineering
 - andrew.silvers@dl-engineering.com

PCB Updates

Layout Updates

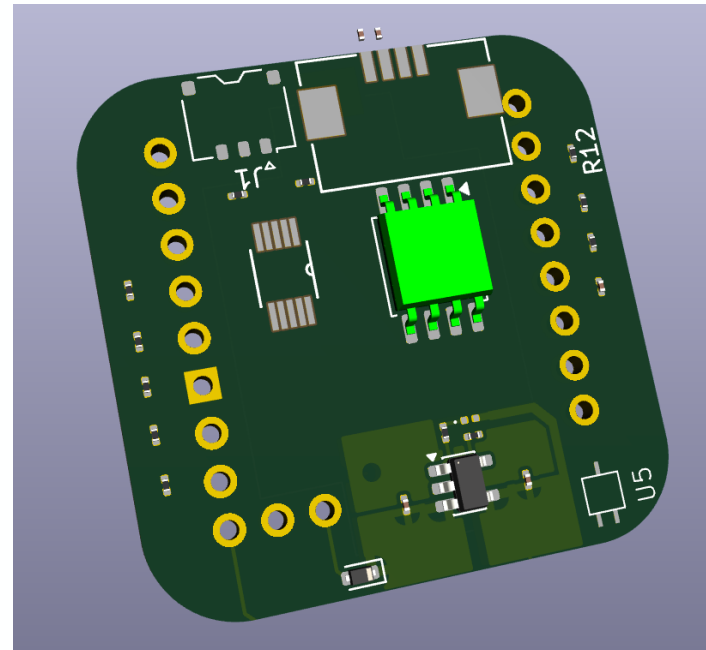
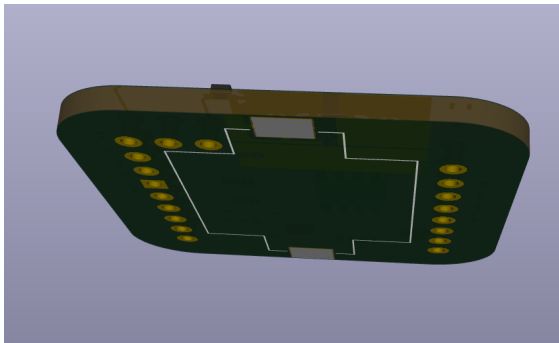
- First board actually had layers flipped
 - Initial oversight, has been corrected
- Height will be 29.180 mm
 - Length of battery holder (squared)
 - Beveled Edges for easier handling



PCB Updates

Design Changes

- Resistors moved to perimeter of device for better heat dissipation
- Memory chip stays in cooler area for sparser track spread-out
- 3D view of device seen below:



IEEE Paper

General Guidelines

- Double checked general guidelines
 - 2 column pages
 - Total of 8 pages maximum
- Completed initial draft
 - Minor questions on formatting
 - Paper seen outside of slides

System Design of a Keel Bone Device

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Abstract—Keel bone damage (KBD) is a prevalent and severe welfare issue in commercial laying hens, often resulting in fractures and deviations that impact both animal well-being and agricultural productivity. Understanding the biomechanical forces and behavioral activities that contribute to these injuries requires in-vivo, real-time monitoring. This paper presents the system design and development of a miniaturized, low-power biomedical printed circuit board (PCB) tailored specifically for keel bone monitoring. The proposed device integrates an inertial measurement unit (IMU) and a low-power microcontroller to capture high-resolution kinetic data. Designed with strict size and weight constraints to avoid altering the animal's natural behavior, the system aims to provide researchers with quantitative data on the physical stressors affecting the keel bone. Preliminary hardware designs, component selection, and power optimization strategies are discussed.

Index Terms—Keel bone, biomedical device, printed circuit board (PCB), animal welfare, wearable sensors, IMU.

I. INTRODUCTION

Keel bone damage (KBD) includes fractures and anatom-

ing high-frequency kinetic data. The system is conceptually divided into sensing, processing, communication, and power management modules.

A. Microcontroller and Wireless Communication

To balance computational requirements with strict power limitations, a System-on-Chip (SoC) featuring an ARM Cortex-M processor integrated with a Bluetooth Low Energy (BLE) radio was selected (Arduino Nicla Sense). This allows the device to process sensor data locally and wirelessly transmit bursts of data to a central receiver in the housing facility, eliminating the need for bulky external wiring.

B. Sensor Integration

Kinetic events are monitored using a 6-axis Inertial Measurement Unit (IMU), comprising a 3-axis accelerometer and a 3-axis gyroscope. The IMU is configured to operate in a low-power, wake-on-motion mode. When an acceleration thresh-

Author Portal:

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IEEE Paper

Documentation for Project

- Refreshed documentation for project
- Other parties (including Nolan can get up to date)
- All on the GitHub page

Biomedical Sensor Node for Keel Bone Monitoring

Overview

This repository contains the hardware design files, firmware, and documentation for a custom, miniaturized Printed Circuit Board (PCB) designed to monitor keel bone dynamics in poultry. Keel bone damage is a significant welfare and economic issue in the commercial egg industry. This biomedical device aims to provide researchers with continuous, high-resolution, in-vivo kinetic data to better understand the biomechanical forces and environmental factors contributing to these injuries.

Key Features

- **Miniaturized Footprint:** Designed with strict size and weight constraints to ensure the device does not alter the natural behavior of the animal.
- **Low-Power Architecture:** Utilizes an event-driven microcontroller and a wake-on-motion Inertial Measurement Unit (IMU) to maximize battery life during longitudinal studies.
- **Wireless Data Transfer:** Integrated Bluetooth Low Energy (BLE) module for seamless, wire-free data transmission to a central receiver in the housing facility.
- **Biocompatible Form Factor:** Hardware is optimized for eventual encapsulation in biocompatible epoxy or medical-grade silicone.

Hardware Specifications

Agriculture Updates

Data working properly

- Spoke with Omosalewa regarding updates and operation
- Data logger working properly, no need to escalate meeting

Agriculture Updates

Data working properly

- Example of data logger with CSV data
- Taken over 16 minutes with several samples
- Data outside of presentation...

Next Week

Plans & Blockers

- Requested data from agriculture department last week
 - Need to do analysis
 - Put results in final paper
- Start insert images and gradual work into paper

Thank You

Purdue Polytechnic Institute



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